PART B — TAPERED WINGS

Airfoil	NACA 2415	**AR**	10	**Velocity (m/s)**	50	**AOA sweep**	to 15°
Method	LLT	**Twist**	0°	**Dihedral**	0°	**Analysis point**	5°

WING GEOMETRY

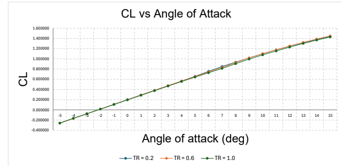
Wing	AR	TR	Root chord (m)	Tip chord (m)	Span (m)	Area (m²)
Wing 1b	10.000	0.200	1.000	0.200	6.000	3.600
Wing 2b	10.000	0.600	1.000	0.600	8.000	6.400
Wing 3b	10.000	1.000	1.000	1.000	10.000	10.000

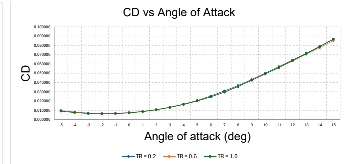
LIFT-CURVE SLOPE

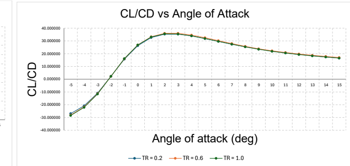
Wing	TR	XFLR5 slope (1/deg)	XFLR5 slope (1/rad)	Theoretical slope (1/rad)
Wing 1b	0.2000	0.0922	5.2850	5.2133
Wing 2b	0.6000	0.0910	5.2145	5.2159
Wing 3b	1.0000	0.0891	5.1064	5.1685

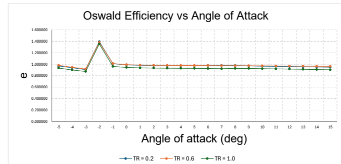
THEORETICAL VALUES AT α = 5°

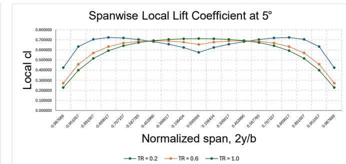
Wing	TR	Total CL	Induced AOA (deg)	Effective AOA (deg)	Downwash magnitude (m/s)	Induced CD
Wing 1b	0.2000	0.6506	1.2174	3.7826	1.0626	0.0138
Wing 2b	0.6000	0.6529	1.2183	3.7817	1.0633	0.0139
Wing 3b	1.0000	0.6475	1.2734	3.7266	1.1114	0.0144

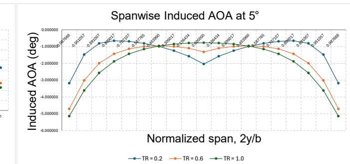


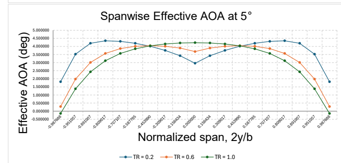


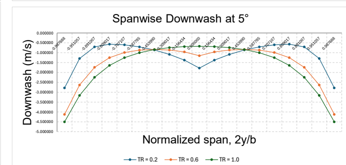


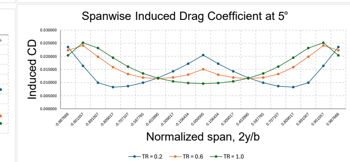












RAW POLAR DATA

α (deg)	CL TR=0.2	CL TR=0.6	CL TR=1.0	CD TR=0.2	CD TR=0.6	CD TR=1.0	CL/	CL/CD TR=0.6	CL/CD TR=1.0	e TR=0.2	e TR=0.6	e TR=1.0
-5	-0.261091	-0.261968	-0.258229	0.009687	0.009330	0.009087	##	-28.076670	-28.418900	0.970101	0.982285	0.933841
-4	-0.168428	-0.169150	-0.166870	0.008078	0.007786	0.007555	##	-21.725950	-22.085950	0.939288	0.947253	0.900006
-3	-0.076638	-0.077364	-0.076636	0.007056	0.006805	0.006588	##	-11.368700	-11.632590	0.906155	0.917030	0.873655
-2	0.015058	0.014395	0.013697	0.006642	0.006417	0.006217	##	2.243130	2.203059	1.393918	1.379624	1.354200
-1	0.106951	0.106439	0.104250	0.006808	0.006617	0.006447	##	16.086870	16.171240	1.010056	1.011966	0.961414
0	0.198587	0.198394	0.194891	0.007555	0.007391	0.007259	##	26.842650	26.847020	0.988442	0.992713	0.944655
1	0.290036	0.290010	0.285122	0.008884	0.008704	0.008615	##	33.319680	33.096680	0.981631	0.985735	0.936658
2	0.381103	0.381416	0.375221	0.010798	0.010600	0.010549	##	35.984070	35.570360	0.977367	0.982137	0.933843
3	0.471768	0.472618	0.465014	0.013363	0.013127	0.013121	##	36.003570	35.441620	0.973815	0.979945	0.931004
4	0.563029	0.563217	0.554110	0.016599	0.016273	0.016314	##	34.611460	33.964370	0.973237	0.978847	0.929075
5	0.657115	0.653131	0.641688	0.020609	0.020065	0.020116	##	32.550040	31.899490	0.974575	0.977416	0.927360
6	0.755272	0.742948	0.727011	0.025491	0.024568	0.024511	##	30.240000	29.660790	0.978137	0.975966	0.924923
7	0.848269	0.831540	0.811301	0.030928	0.029757	0.029556	##	27.944730	27.449600	0.977938	0.973790	0.923134
8	0.933117	0.924890	0.903752	0.036669	0.035896	0.035652	##	25.765850	25.349440	0.975803	0.973635	0.926338
9	1.018005	1.017666	0.994351	0.043153	0.042690	0.042376	##	23.838250	23.464930	0.971285	0.973264	0.926172
10	1.099869	1.101037	1.076547	0.050077	0.049454	0.049237	##	22.263870	21.864490	0.966201	0.971694	0.922042
11	1.176383	1.175298	1.153074	0.057093	0.056044	0.056175	##	20.970970	20.526380	0.962452	0.969311	0.918306
12	1.247416	1.250128	1.228653	0.064118	0.063097	0.063508	##	19.812750	19.346500	0.959761	0.968031	0.915334
13	1.315177	1.322115	1.300175	0.071418	0.070364	0.071021	##	18.789630	18.306900	0.956594	0.966992	0.912097
14	1.378698	1.387116	1.366348	0.078975	0.077617	0.078677	##	17.871200	17.366540	0.952751	0.964828	0.907593
15	1.437607	1.448538	1.428669	0.086820	0.085083	0.086509	##	17.024930	16.514700	0.948683	0.962673	0.903501

SPANWISE DATA AT α = 5°

y TR=0.2	2y/b	Local cl	Induced AOA (°)	Effective AOA (°)	Downwash (m/s)	Induced CD
-2.963065	-0.987688	0.423985	-3.183363	1.816637	-2.780870	0.023557
-2.853170	-0.951057	0.633035	-1.480785	3.519215	-1.292516	0.016361
-2.673020	-0.891007	0.705529	-0.806930	4.193070	-0.704226	0.009936
-2.427051	-0.809017	0.723695	-0.652074	4.347926	-0.569066	0.008236
-2.121320	-0.707107	0.718899	-0.687126	4.312874	-0.599660	0.008621
-1.763356	-0.587785	0.703289	-0.808256	4.191744	-0.705384	0.009921
-1.361971	-0.453990	0.682348	-0.988487	4.011514	-0.862703	0.011772
-0.927051	-0.309017	0.656888	-1.245480	3.754520	-1.087058	0.014279
-0.469303	-0.156434	0.624589	-1.575587	3.424413	-1.375306	0.017176
0.000000	0.000000	0.575758	-2.038199	2.961801	-1.779415	0.020482
0.469303	0.156434	0.624589	-1.575587	3.424413	-1.375306	0.017176
0.927051	0.309017	0.656888	-1.245480	3.754520	-1.087058	0.014279
1.361971	0.453990	0.682348	-0.988487	4.011514	-0.862703	0.011772
1.763356	0.587785	0.703289	-0.808256	4.191744	-0.705384	0.009921
2.121320	0.707107	0.718899	-0.687126	4.312874	-0.599660	0.008621
2.427051	0.809017	0.723695	-0.652074	4.347926	-0.569066	0.008236
2.673020	0.891007	0.705529	-0.806930	4.193070	-0.704226	0.009936
2.853170	0.951057	0.633035	-1.480785	3.519215	-1.292516	0.016361
2.963065	0.987688	0.423985	-3.183363	1.816637	-2.780870	0.023557

y TR=0.6	2y/b	Local cl	Induced AOA (°)	Effective AOA (°)	Downwash (m/s)	Induced CD
-3.950753	-0.987688	0.270937	-4.714743	0.285257	-4.123701	0.022295
-3.804226	-0.951057	0.458037	-3.016372	1.983628	-2.634716	0.024114
-3.564026	-0.891007	0.569963	-1.992495	3.007505	-1.739481	0.019821
-3.236068	-0.809017	0.633720	-1.432650	3.567350	-1.250484	0.015846
-2.828427	-0.707107	0.668207	-1.133905	3.866095	-0.989648	0.013224
-2.351141	-0.587785	0.685076	-0.993605	4.006395	-0.867171	0.011880
-1.815962	-0.453991	0.690860	-0.953810	4.046190	-0.832433	0.011501
-1.236068	-0.309017	0.688172	-0.992002	4.007998	-0.865771	0.011915
-0.625738	-0.156434	0.677591	-1.098413	3.901587	-0.958664	0.012990
0.000000	0.000000	0.654052	-1.319659	3.680341	-1.151823	0.015064
0.625738	0.156434	0.677591	-1.098413	3.901587	-0.958664	0.012990
1.236068	0.309017	0.688172	-0.992002	4.007998	-0.865771	0.011915
1.815962	0.453991	0.690860	-0.953810	4.046190	-0.832433	0.011501
2.351141	0.587785	0.685076	-0.993605	4.006395	-0.867171	0.011880
2.828427	0.707107	0.668207	-1.133905	3.866095	-0.989648	0.013224
3.236068	0.809017	0.633720	-1.432650	3.567350	-1.250484	0.015846
3.564026	0.891007	0.569963	-1.992495	3.007505	-1.739481	0.019821
3.804226	0.951057	0.458037	-3.016372	1.983628	-2.634716	0.024114
3.950753	0.987688	0.270937	-4.714743	0.285257	-4.123701	0.022295

y TR=1.0	2y/b	Local cl	Induced AOA (°)	Effective AOA (°)	Downwash (m/s)	Induced CD
-4.938442	-0.987688	0.226872	-5.143482	-0.143482	-4.500631	0.020366
-4.755283	-0.951057	0.398835	-3.617828	1.382172	-3.161353	0.025184
-4.455033	-0.891007	0.516065	-2.573011	2.426989	-2.246886	0.023175
-4.045085	-0.809017	0.592676	-1.883701	3.116299	-1.644432	0.019485
-3.535534	-0.707107	0.641667	-1.433478	3.566522	-1.251207	0.016054
-2.938926	-0.587785	0.672941	-1.146072	3.853928	-1.000270	0.013461
-2.269952	-0.453990	0.692637	-0.963599	4.036401	-0.840978	0.011649
-1.545085	-0.309017	0.704464	-0.850358	4.149642	-0.742132	0.010455
-0.782172	-0.156434	0.710797	-0.789718	4.210282	-0.689203	0.009797
0.000000	0.000000	0.712796	-0.770568	4.229432	-0.672488	0.009586
0.782172	0.156434	0.710797	-0.789718	4.210282	-0.689203	0.009797
1.545085	0.309017	0.704464	-0.850358	4.149642	-0.742132	0.010455
2.269952	0.453990	0.692637	-0.963599	4.036401	-0.840978	0.011649
2.938926	0.587785	0.672941	-1.146072	3.853928	-1.000270	0.013461
3.535534	0.707107	0.641667	-1.433478	3.566522	-1.251207	0.016054
4.045085	0.809017	0.592676	-1.883701	3.116299	-1.644432	0.019485
4.455033	0.891007	0.516065	-2.573011	2.426989	-2.246886	0.023175
4.755283	0.951057	0.398835	-3.617828	1.382172	-3.161353	0.025184
4.938442	0.987688	0.226872	-5.143482	-0.143482	-4.500631	0.020366